

Programme Code: TU256/TU059/TU060
Module Code: SPEC 9270

TECHNOLOGICAL UNIVERSITY DUBLIN

Grangegorman

TU256 - POSTGRADUATE CERTIFICATE IN DATA SCIENCE

TU059/TU060 - MSc IN COMPUTER SCIENCE

Year 1/2

SEMESTER 2 EXAMINATIONS 2022/23

SPEC 9270 Machine Learning

Internal Examiners:

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Instructions to Candidates: ANSWER QUESTION 1 AND 2 OTHER QUESTIONS

Duration: 2 HOURS

Special Instructions/Handouts/Materials Required: NONE

1. (a) Explain what is meant by a model being *consistent* with the training data. (4 marks)
Explain when a model might not be consistent with the training data.
- (b) Describe two causes of *overfitting* in supervised machine learning. (4 marks)
- (c) Data leakage can give over-optimistic model performance. Explain why and discuss what is necessary to avoid data leakage. (4 marks)
- (d) When decision tree pruning results in the removal of a sub-tree with the root of the sub-tree becoming a leaf node, what is the class label associated with this new leaf node? (4 marks)
- (e) Is the following statement true or false? Explain your answer. (4 marks)
“A *k*-Nearest Neighbour classifier will always have perfect accuracy if the training data is used for testing”
- (f) In error based learning, explain what an error or loss function is and give an example. (4 marks)
- (g) A linear regression model to predict drug dosage (in mg) has returned feature weights of 0.1 and 0.03 for features *weight* and *height* respectively along with an intercept of -4.3. What will the model predict for a 60kg and 165cm person? (4 marks)
- (h) In machine learning what is the difference between hyperparameters and other parameters? Give an example of each. (4 marks)
- (i) Why is it not possible to use *k*-Means clustering with categorical data? (4 marks)

2. (a) You have been asked to build a classifier that will predict whether an individual should qualify for a credit card or not. You have been provided with lists of individuals who have already applied for credit cards along with their outcome, whether they were awarded a credit card or not. You have been told that the key characteristics of the individual that are typically used to determine the outcome include Age, their employment status (i.e. employed or not), and whether they are a home owner or not. Describe how you will build a Naive Bayes classifier which can be used to predict the outcome. (10 marks)
- (b) i. If you have a collection of 3 green balls and 5 red balls and you add 2 green balls to the collection what happens to the entropy of the collection? (4 marks)
- ii. Entropy is used in training a decision tree. Explain how it is used. (6 marks)
- (c) Naive Bayes classifiers and decision trees are eager learners, explain what is meant by the term “eager learner”. What is the difference between an eager and a lazy learner? Give an example of a lazy learner and discuss a scenario where a lazy learner may be more beneficial than an eager learner. (12 marks)

3. (a) What is the role for confusion matrices in the evaluation of machine learning classifiers? What insights might be offered from examining a confusion matrix that might not come from considering the misclassification rate only? (8 marks)
- (b) What is the difference between precision and recall? Discuss one scenario when precision may be more appropriate to use than recall. (7 marks)
- (c) An important principle in evaluating machine learning models is that the test data should be accessed in the model training process because it can result in unrealistic estimate of generalisation accuracy. This applies to the data pre-processing pipeline as well as the model fitting itself. In practice, some scenarios are more serious than others; comment on the seriousness of each of the following scenarios: (9 marks)
- (i) The test data is included during the feature selection process.
 - (ii) The test data is included when fitting a One-Hot-Encoder
 - (iii) Missing values are replaced using the mean values for features across both training and test sets, i.e. the test data is used when the means are being calculated
- (d) When compared with hold-out testing, cross validation testing can provide estimates of generalisation accuracy that are (i) more accurate and (ii) more stable. Explain both of these claims. (8 marks)

4. (a) i. The *curse of dimensionality* in machine learning is one reason to reduce the dimensionality in a dataset. Explain the *curse of dimensionality* and give two other benefits of reducing dimensionality. (7 marks)
- ii. You have been asked to train a model on a text dataset that has high dimensionality and is sparse. Explain how you would reduce the dimensionality of the dataset and identify the best subset of features to use. (9 marks)
- (b) i. Diversity is important in ensemble learning. Bagging (bootstrap aggregation) has a mechanism for achieving diversity; explain how it works. (6 marks)
- ii. Bagging will not produce diversity in ensembles of nearest-neighbour classifiers; describe a process that will. (4 marks)
- iii. Explain what differentiates the different ensemble members in a boosting ensemble. (6 marks)